

Mikael Ade Krisna Respati

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Profile

AI Engineer with a strong background in **machine learning**, **deep learning**, and **data-driven system design**. Experienced in developing **end-to-end AI pipelines**, implementing **neural network architectures** (CNN, LSTM, Transformer, Autoencoder), and optimizing models for **real-world deployment**. Skilled in **Python**, **TensorFlow**, **PyTorch**, and data preprocessing, with hands-on experience integrating AI into practical applications such as **prediction systems**, **intelligent sensing**, and **embedded devices**. Passionate about building **robust, scalable, and efficient AI solutions** that drive automation and intelligent decision-making across diverse domains.

Education

Sejong University, Seoul, South Korea

Feb 2024 – Feb 2026 (*Expected*)

Master's in AI and Robotics, GPA: 4.20 / 4.50

Thesis: Joint Beamforming and Localization using Multi-Task Deep Learning

Developed a CNN-based model using a soft parameter sharing approach with attention gates and designed an adaptive filter for CSI dimensionality reduction, trained end-to-end with the model.

Gadjah Mada University, Yogyakarta, Indonesia

Aug 2018 – Nov 2022

Bachelor of Electrical Engineering, GPA: 3.20 / 4.00

Thesis: Development of Indirect Mass Reporting System through Twitter Based on NLP

Developed an SVM-based ML model to classify tweets as public service complaints or non-complaints in the Yogyakarta region.

Experience

National Taiwan University of Science and Technology (NTUST), Taipei, Taiwan

Jul 2023 – Nov 2023

Research Assistant Intern, Center for IoT Innovation (CITI)

Developed a Fault Prediction System for Wind Turbine Generators using LSTM networks in Python (TensorFlow, scikit-learn). Addressed data imbalance with SMOTEEN and implemented an Autoencoder with Attention layers, achieving MAE = 1.36 and $R^2 = 0.9$.

PT. Lunar Inovasi Teknologi, Yogyakarta, Indonesia

Dec 2022 – May 2023

Hardware Test Engineer

Designed and developed PCBs for power monitoring ICs (ADE9153A and RN8902C) integrated with an ESP32-S3 microcontroller and OLED display to create a Smart Power Metering product using Altium Designer.

PT. Lunar Inovasi Teknologi, Yogyakarta, Indonesia

Sep 2022 – Nov 2022

Internship Program

Designed and developed a power monitoring minimum system board utilizing RN8209C, V9021, and V9021S ICs using Altium Designer.

Publications

IEEE Access (2024): *A Survey on Machine Learning Enhanced Integrated Sensing and Communication Systems: Architectures, Algorithms, and Applications.*

Skills & Certifications

Technical: Python, C/C++, MATLAB, CNN, LSTM, Autoencoder, Attention Mechanism, AI Agents, ESP32, PCB Design (Altium, Eagle), Microcontrollers

Certificates: DeepLearning.AI TensorFlow Developer Specialization (2021)

View Certificate

Languages: English, Bahasa Indonesia